

# QUOC HUY DINH (RYAN)

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## PROFESSIONAL SUMMARY

Artificial Intelligence and Computer Science student at the University of Technology Sydney (UTS), recognised on the FEIT Dean's List 2026 for outstanding academic achievement. Hands-on experience building AI-powered systems, deep learning models from scratch, Python data pipelines, and full-stack web applications. Currently a Research Assistant working on Vision-Language Models and generative modelling, leading the ML side of a computer-vision research project targeting international publication, and preparing a solo research submission for DICTA 2026. Comfortable taking work end to end, from research and data preprocessing through to shipped products. Open to internship, casual, part-time, contract, and graduate roles across AI, software, and data. Fast learner who delivers strong outcomes both independently and in teams.

## EDUCATION

**Bachelor of Artificial Intelligence** University of Technology Sydney (UTS) | Sydney, Australia | Jul 2025 to Present

- Focus on machine learning, deep learning, and applied AI systems
- Database Fundamentals: High Distinction (100/100), full marks across relational algebra, joins, aggregation, and query optimisation

**Bachelor of Computer Science** Ho Chi Minh City University of Technology (HCMUT) | Aug 2023 to Jun 2025

- GPA: 3.6 / 4.0
- Data Structures and Algorithms (C++): A grade

**Le Hong Phong High School for the Gifted** Specialised Physics Class | Ho Chi Minh City | 2020 to 2023

- Top-ranked specialised high school for STEM in Ho Chi Minh City
- Physics-major stream covering advanced classical mechanics, electromagnetism, and quantum physics

## EXPERIENCE

**Research Assistant** University of Technology Sydney | Sydney, NSW | Nov 2025 to Present

- Cleaned and analysed noisy multi-source Optical Character Recognition (OCR) datasets, ran iterative model experiments, and translated findings into reproducible workflows
- Built normalisation, cleaning, and feature-extraction pipelines for downstream modelling across multi-source datasets
- Contributed backend server modules (TCP, OBS WebSocket, and REST API layers) for a game-research project, enabling real-time messaging and programmatic control
- Currently researching Vision-Language Models (VLM) and Generative Adversarial Networks (GAN), running multimodal understanding and generative modelling experiments

## AI AND MACHINE LEARNING PROJECTS

**Transformer Chatbot from Scratch** PyTorch, Lightning, FastAPI, Hugging Face Spaces

- Built a decoder-only transformer entirely from scratch, including a custom tokenizer, causal masking, and temperature sampling, without relying on high-level model libraries
- Architecture: 4 decoder blocks, d\_model 256, 4 attention heads, word-level vocabulary trained on a 40-intent dataset
- Served the model through a FastAPI backend and deployed it on Hugging Face Spaces, powering a live chatbot widget on a personal portfolio site

**Smart Parking and Intelligent Traffic Management (ALPR)** ML Team Lead | PyTorch, Ultralytics YOLO, PARSEQ, ByteTrack, Streamlit

- Leading the machine learning work on a full automatic licence-plate recognition pipeline targeting international publication
- Combined YOLOv11-OBb plate detection, a PARSEQ transformer OCR head, ByteTrack temporal voting, and Zero-DCE++ low-light enhancement
- Benchmarked robustness across CCPD2020 subsets (base, blur, tilt, rain, night, challenge) and built a Streamlit dashboard for real-time violation detection
- Team lead for a group of three across five ablation configurations spanning detector and OCR axes

**Hybrid CV plus VLM for Tactical Badminton Analysis** Solo research, target DICTA 2026 (deadline 1 Jul 2026) | PyTorch, Ultralytics

YOLOv11, Qwen2.5-VL, Gemini, LangChain, FAISS, Streamlit

- Designing a hybrid pipeline that decomposes tactical badminton analysis into a geometric side (YOLOv11 court-keypoint homography, player and shuttle tracking, Y-reversal shot detection) and a semantic side (a Vision-Language Model handling shot type, handedness, and out-of-position state)
- Achieved court keypoint mAP 0.95, player detection mAP 0.99, and shuttle mAP 0.57; shot-trigger v2 (Y-reversal batch detector) reaches F1 0.80 on a 30-second Ginting-Axelsen rally with 30 hand-annotated ground-truth shots
- Wired shot events into a conversational RAG interface (LangChain plus FAISS plus Gemini 2.5 Pro) over a static rules-and-tactics knowledge base, served through a Streamlit UI
- Core contribution: task-decomposition ablation (CV-only vs VLM-only vs Hybrid) evaluated against hand-annotated ground truth from professional BWF rallies

**Machine Learning Models from Scratch** Python, NumPy

- Implemented regression and classification algorithms from first principles, applying gradient descent and regularisation, and analysed the bias-variance trade-off and convergence behaviour

## SOFTWARE ENGINEERING PROJECTS

**Personal Portfolio Site (this website)** Next.js 16, TypeScript, Tailwind CSS, Framer Motion, Resend, FastAPI

- Built a multi-page Next.js portfolio (App Router) with shared layout for navbar, footer, and a chatbot widget, plus dedicated routes for resume, projects, research, and contact
- Integrated a Resend-powered contact form with server-side validation, and proxied chatbot requests through a Next.js API route to a separate FastAPI service deployed on Hugging Face Spaces
- Implemented dark mode by default, responsive layout, route-based active navigation, and project filtering by industry
- Live in production at [cv-website-lemon.vercel.app](https://cv-website-lemon.vercel.app), deployed on Vercel with CI on every push to main

### **AI-Powered Coffee Shop Manager Dashboard** Python, LangChain, Redis, Docker, AWS (planned)

- Designed and built an end-to-end AI briefing system delivering automated morning reports covering revenue trends and operational insights, removing the need for manual data review
- Developed a single LangChain pipeline integrating four AI capabilities: natural language Q&A, revenue summarisation, inventory forecasting, and scheduled auto-generated daily reports
- Implemented a Redis caching layer to serve repeated queries without redundant recomputation, and containerised the system with Docker for consistent deployment

### **CurricuLLM, Parent-Teacher AI Platform** Cambridge EduX Hackathon 2026, Team of 4 | TypeScript, React, FastAPI, PostgreSQL

- Built an AI pipeline that auto-generates personalised curriculum progress reports for parents from teacher input, removing manual report writing
- Developed a parent-facing AI chat interface for natural language queries on student progress, served via FastAPI, with real-time multilingual translation
- Contributed full-stack across the React frontend and the FastAPI/PostgreSQL backend (REST API design, database schema)

### **Game Automation and Streaming Integration Tool** Python, Flask, FastAPI, TCP, OBS WebSocket, REST API

- Implemented a TCP layer for real-time client-server communication across two repositories (main app and standalone TCP server/client), enabling low-latency bidirectional messaging
- Integrated the OBS WebSocket protocol to react to live stream events and exposed keyboard/mouse triggers via REST API endpoints across a modular backend
- Designed the boundary between control plane and event plane so the OBS integration and the keyboard/mouse triggers can be extended independently

### **E-Commerce Web Application** React, Express.js, Node.js, PostgreSQL, JWT

- Built a full-stack e-commerce platform independently, covering product listing, search and filter, shopping cart, and checkout with order management
- Implemented JWT-based authentication (register, login, protected routes) with bcrypt password hashing
- Designed and normalised a relational PostgreSQL schema (up to 3NF) and optimised queries for CRUD operations

### **Frontend Projects** React, JavaScript, HTML, CSS

- Built a React Tic-Tac-Toe game with move history and undo (deployed on Netlify), a React Music Player, and a React Weather App
- Built pure HTML/CSS/JS UI clones (Steam-style marketplace and job-search interfaces) focused on responsive layout and accessible markup

## **DATA AND ANALYTICS PROJECTS**

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### **Game Telemetry Data Analysis** Python, Pandas, NumPy, Matplotlib, Seaborn

- Ran exploratory data analysis on competitive-game telemetry: cleaning, feature engineering, and statistical analysis of player behaviour and match outcomes, surfacing actionable performance and outlier insights

### **OCR Preprocessing Pipeline** Python, OpenCV, NumPy, PaddleOCR

- Built an image preprocessing pipeline (adaptive thresholding, denoising, deskewing, contrast enhancement) and benchmarked downstream OCR accuracy across configurations to find the strongest pipeline for noisy inputs

### **Housing Price Data Analysis** Python, Pandas, scikit-learn, Matplotlib

- Performed EDA, missing-value handling, and statistical modelling to identify the strongest predictors of house price, using hypothesis testing, correlation analysis, and regression diagnostics

## **CERTIFICATIONS**

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### **WIL to Work Certificate of Completion** UTS Careers | 2026

- Work Integrated Learning preparation programme delivered by UTS Careers, covering preparation for WIL experiences, succeeding in WIL activities, Australian workplace cultures, and undertaking WIL in online environments
- Signed by Eva Chan, Manager, UTS Careers

### **Full-Stack Web Development** CoderSchool | 2024

- Intensive full-stack programme covering React, Node.js/Express, MongoDB/PostgreSQL, REST API design, and deployment workflows

## **AWARDS**

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### **Dean's List 2026** UTS Faculty of Engineering and Information Technology | Faculty level, university | 2026

- Included on the FEIT Dean's List for outstanding academic achievement, a faculty initiative celebrating top student performance in Engineering and Information Technology
- Certified by Professor Karen Whelan, Associate Dean (Teaching and Learning), Faculty of Engineering and Information Technology

### **Second Prize, Physics Olympiad** Ho Chi Minh City Department of Education | City level, high school | 2022

- Awarded while studying at Le Hong Phong High School for the Gifted

### **Second Prize, Scientific Research Contest** Ho Chi Minh City Department of Education | City level, middle school | 2018

## **SKILLS**

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**Programming Languages:** Python, C++, JavaScript, TypeScript, HTML/CSS

**AI / Machine Learning:** PyTorch, PyTorch Geometric, Lightning, scikit-learn, LangChain, Ultralytics YOLO, Natural Language Processing (NLP), Large Language Models (LLM), Vision-Language Models (VLM), GANs, Transformers, Data Cleaning, Feature Extraction

**Frameworks and Libraries:** Next.js, React, Tailwind CSS, Flask, FastAPI, Express.js, Node.js, Pandas, NumPy, OpenCV, Streamlit, Matplotlib, Seaborn

**Databases:** PostgreSQL, Redis

**Tools and Platforms:** Vercel, Docker, Git, Jupyter, VS Code, Hugging Face Spaces, AWS (planned)

**Protocols and Integration:** TCP/IP, OBS WebSocket, REST APIs, JWT Authentication, Resend